0.1 重积分计算

例题 0.1 设 $f(x,y)$ 在 D 上有定义, 且在某一圆盘外为 0. 求证:

$$\iint_D \left[\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2\right] \mathrm{d}x\mathrm{d}y = 0.$$

证明 Show/Hide Proof

由题设可知 f 在 ∂D 上恒为 0, 进而

$$\frac{\partial f}{\partial x}(x,y) = \frac{\partial f}{\partial y}(x,y) = 0, \quad \forall (x,y) \in \partial D.$$

由 Green 公式可得

$$\begin{aligned}
\iint_D \left[\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2\right] \mathrm{d}x\mathrm{d}y &= \iint_D (f_{xx}f_{yy} - f_{xy}^2)\mathrm{d}x\mathrm{d}y \\
&= \iint_D \left[(f_{xx}f_y)_y - (f_{xy}f_y)_x\right] \mathrm{d}x\mathrm{d}y \\
&\xlongequal{\text{Green 公式}} \oint_{\partial D} f_{xy}f_y\mathrm{d}x + f_{xx}f_y\mathrm{d}y = 0.
\end{aligned}$$

□